

NOVYY USHTAGAN, Kazakhstan

WMO: 346910

Lat: 47.9056N

Lon: 48.7906E

Elev: -33

StdP: 14.71

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.4	(c) -3.8	(d) -15.5	(e) 2.4	(f) -8.6	(g) -9.9	(h) 3.2	(i) -2.7	(j) 22.5	(k) 20.6	(l) 19.9	(m) 22.4	(n) 5.6	(o) 320	(p) 0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 22.2	(c) 100.5	(d) 67.9	(e) 97.5	(f) 67.2	(g) 94.3	(h) 65.9	(i) 70.8	(j) 92.6	(k) 69.2	(l) 90.6	(m) 67.8	(n) 89.0	(o) 7.3	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	89.7	76.1	62.1	83.4	76.0	60.1	77.4	76.4	34.5	92.2	33.2	90.7	32.1	89.3	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.1	(b) 18.6	(c) 17.1	(e) -13.2	(f) 104.6	(g) 8.7	(h) 2.5	(i) -19.5	(j) 106.3	(k) -24.6	(l) 107.8	(m) -29.5	(n) 109.2	(o) -35.8	(p) 111.0		
(a) 22.1	(b) 18.6	(c) 17.1	(e) -13.7	(f) 73.5	(g) 8.4	(h) 2.4	(i) -19.8	(j) 75.3	(k) -24.7	(l) 76.7	(m) -29.5	(n) 78.1	(o) -35.6	(p) 79.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	20.1	21.9	35.7	52.3	66.0	75.3	79.7	77.0	63.8	49.2	33.9	24.5	
	DBStd	23.24	11.28	12.15	8.40	8.64	8.56	7.24	5.95	7.48	8.05	8.68	10.41	10.38	
	HDD50	3637	927	787	446	73	3	0	0	0	4	121	485	791	
	HDD65	6797	1392	1207	908	389	93	9	0	5	114	492	932	1256	
	CDD50	3680	0	0	3	142	499	758	922	837	419	97	3	0	
	CDD65	1364	0	0	0	8	123	316	457	378	79	3	0	0	
	CDH74	17186	0	0	0	129	1480	3867	5836	4893	946	35	0	0	
	CDH80	8712	0	0	0	24	610	1944	3155	2648	327	4	0	0	
Wind	WSAvg	6.8	6.9	7.7	7.9	7.5	7.2	6.4	5.8	5.9	6.2	6.3	6.4	7.0	
Precipitation	PrecAvg	7.8	0.6	0.5	0.6	0.7	0.9	0.9	0.5	0.4	0.6	0.8	0.6	0.7	
	PrecMax	11.8	1.8	2.0	2.0	1.5	2.8	2.3	2.0	1.4	1.8	3.4	1.5	1.5	
	PrecMin	5.5	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	PrecStd	1.7	0.4	0.4	0.5	0.5	0.6	0.6	0.6	0.4	0.5	0.7	0.4	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	46.8	62.8	83.4	95.0	102.2	104.8	103.5	92.9	78.1	59.5	46.2	
		MCWB	40.6	41.7	49.0	59.7	62.9	66.8	68.5	68.9	64.2	59.3	51.2	42.7	
	2%	DB	39.2	41.4	57.3	77.7	90.7	98.0	100.6	100.1	87.7	72.9	54.6	42.8	
		MCWB	37.1	37.3	46.1	57.5	62.7	66.6	68.0	68.7	62.8	56.5	49.3	40.7	
	5%	DB	35.9	38.2	52.3	72.9	87.1	94.4	97.6	96.9	83.6	67.9	50.7	39.4	
		MCWB	33.9	35.0	43.6	55.4	61.6	65.6	67.4	67.5	60.5	54.1	46.5	37.0	
	10%	DB	33.5	35.7	47.9	68.2	82.6	90.6	93.9	93.2	79.4	63.2	47.2	36.5	
		MCWB	31.7	33.2	40.9	53.1	60.3	64.3	66.1	66.0	59.1	52.9	43.8	34.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.7	42.2	50.5	61.5	67.3	71.5	72.7	73.3	67.4	62.0	53.8	43.8
			MCDB	42.9	46.0	59.2	77.7	87.7	93.0	95.6	95.7	85.4	73.6	56.8	45.2
2%		WB	37.2	38.2	47.0	58.8	65.0	69.4	70.7	70.8	64.8	58.5	50.2	40.5	
		MCDB	38.9	40.7	55.7	74.8	83.4	89.5	92.6	93.1	81.2	68.0	53.6	42.4	
5%		WB	34.1	35.5	44.2	56.6	63.2	67.5	69.2	69.0	62.7	56.0	47.0	37.4	
		MCDB	35.7	37.6	51.5	70.4	81.3	87.5	90.7	91.2	78.7	65.4	49.9	39.2	
10%		WB	32.0	33.4	41.5	54.3	61.4	66.0	67.9	67.4	60.7	53.3	43.9	34.6	
		MCDB	33.4	35.4	47.1	66.4	79.3	85.4	88.9	89.3	75.7	62.4	47.2	36.2	
Mean Daily Temperature Range		5% DB	MDBR	11.8	14.1	15.9	21.1	21.9	22.2	22.2	24.1	22.9	19.4	13.3	10.5
			MCDBR	9.6	13.6	22.8	27.5	27.1	26.9	26.5	28.2	28.8	26.5	17.6	9.9
	MCWBR		8.4	10.0	13.4	12.7	10.3	9.6	9.0	10.2	12.0	13.4	11.9	8.3	
	5% WB	MCDBR	8.9	11.3	20.9	25.3	23.7	23.1	23.5	24.9	23.8	22.8	15.3	9.6	
		MCWBR	8.2	9.1	13.1	12.9	10.6	9.6	9.4	10.3	11.5	13.4	11.6	8.5	
Clear-Sky Solar Irradiance	taub	0.350	0.373	0.410	0.474	0.453	0.425	0.431	0.441	0.403	0.375	0.354	0.351		
	taud	2.093	2.048	2.081	1.961	2.094	2.244	2.238	2.182	2.238	2.257	2.217	2.109		
	Ebn at Noon	213	238	251	246	258	266	263	254	252	238	213	195		
	Edh at Noon	29	38	43	54	49	43	42	43	37	30	26	26		
All-Sky Solar Radiation	RadAvg	342	657	1053	1524	1948	2128	2079	1828	1367	829	412	245		
	RadStd	75	111	104	111	146	170	128	108	98	87	61	45		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	+2.74	N/A	-1.31	N/A	N/A	+306	+207

Nomenclature: See separate page